

STANDARD OPERATING PROCEDURE

SOP-MAINT-017: Centrifugal Pump Maintenance

Revision: 4.2

Effective Date: 2025-01-15

Approved by: Chief Maintenance Engineer

Scope: All centrifugal pumps in CDU, VDU, and FCC units.

1. PURPOSE

This SOP establishes mandatory maintenance procedures for centrifugal pumps in refinery service, including inspection criteria, corrective action thresholds, and documentation requirements.

2. VIBRATION MONITORING CRITERIA

2.1 All centrifugal pumps shall be monitored per ISO 10816-7.

2.2 Vibration severity thresholds for pumps rated above 15 kW:

Zone A (Good): < 3.5 mm/s RMS

Zone B (Acceptable): 3.5 - 7.1 mm/s RMS

Zone C (Alert): 7.1 - 11.0 mm/s RMS · Plan corrective action

Zone D (Danger): > 11.0 mm/s RMS · STOP IMMEDIATELY, mandatory inspection before restart.

2.3 When vibration exceeds Zone C threshold (7.1 mm/s):

- a) Notify shift supervisor within 1 hour.
- b) Initiate root cause analysis.
- c) Schedule corrective maintenance within 1 week.

2.4 When vibration exceeds Zone D threshold (11.0 mm/s):

- a) STOP pump immediately (emergency shutdown if necessary).
- b) Notify plant manager and safety department.
- c) Do not restart without documented inspection and clearance.
- d) Perform comprehensive vibration spectrum analysis.
- e) Check bearing condition, rotor balance, and coupling alignment.

2.5 Sub-synchronous vibration below 0.5x running speed indicates potential oil whirl/whip in journal bearings. This requires immediate investigation of bearing oil supply and clearances.

3. MECHANICAL SEAL MAINTENANCE

3.1 Mechanical seal inspection shall be performed:

- a) During every pump overhaul.
- b) When visible leakage exceeds 10 drops per minute.
- c) When process fluid contamination is suspected.

3.2 Seal replacement criteria:

- a) Visible leakage rate > 10 drops/min · REPLACE within 48 hours.
- b) Seal face scoring or chipping · REPLACE immediately.
- c) Carbon ring wear > 2mm · REPLACE during next shutdown.

3.3 Seal replacement procedure:

- Step 1: Isolate pump (LOTO per SOP-SAFE-003).
- Step 2: Drain and flush seal chamber.
- Step 3: Remove old seal assembly.
- Step 4: Inspect shaft and seal cavity for damage.
- Step 5: Verify shaft runout < 0.05 mm.
- Step 6: Install new seal per manufacturer specifications.
- Step 7: Verify seal alignment using laser alignment tool.
- Step 8: Pressure test seal chamber.
- Step 9: Document all measurements in maintenance record.

3.4 Seal misalignment indicators:

- Uneven wear pattern on seal face.
- Carbon deposits on atmospheric side.
- Excessive leakage with new seal installed.

4. PIPE WALL THICKNESS & CORROSION

4.1 Minimum wall thickness requirements:

- a) All piping shall maintain thickness above the calculated minimum retirement thickness per API 570.
- b) For carbon steel pipes in hydrocarbon service:
Minimum retirement thickness = design thickness - corrosion allowance.

4.2 Corrosion monitoring frequency:

- a) Corrosion rate < 0.1 mm/year: inspect every 5 years.
- b) Corrosion rate 0.1 - 0.25 mm/year: inspect every 2 years.
- c) Corrosion rate 0.25 - 0.5 mm/year: inspect annually.
- d) Corrosion rate > 0.5 mm/year: inspect every 6 months, plan replacement.

4.3 When measured thickness reaches retirement threshold:

- a) Remove pipe section from service.
- b) Replace with approved material specification.
- c) Investigate root cause of accelerated corrosion.

4.4 External corrosion under insulation (CUI):

- a) Inspect all insulated piping operating between 50-175°C.
- b) Remove insulation at suspected locations for visual inspection.
- c) Apply approved coating system after repair.

4.5 Corrosion rate calculation:

Rate = (Original thickness - Measured thickness) / Service years.
Record all measurements in the corrosion monitoring database.

5. BEARING TEMPERATURE MONITORING

5.1 Bearing housing temperature limits:

- a) Normal operating range: $< 75^{\circ}\text{C}$.
- b) Alert level: $75\text{-}90^{\circ}\text{C}$ · investigate cause.
- c) Alarm level: $90\text{-}100^{\circ}\text{C}$ · plan immediate shutdown.
- d) Trip level: $> 100^{\circ}\text{C}$ · automatic or manual emergency shutdown.

5.2 When bearing temperature exceeds alert level (75°C):

- a) Check lubrication oil level and quality.
- b) Verify cooling water supply to bearing housing.
- c) Check for abnormal vibration simultaneously.
- d) Document findings and report to maintenance supervisor.

5.3 When bearing temperature exceeds alarm level (90°C):

- a) Prepare for controlled shutdown.
- b) Notify operations and safety.
- c) Switch to standby pump if available.
- d) Do NOT restart until bearing has been inspected.

5.4 Combined high vibration and high bearing temperature indicates potential bearing failure. This combination requires IMMEDIATE shutdown regardless of individual threshold levels.